



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT (Part II)

Date: 01.08.2026

In addition to Part I (the General Handout for all courses, appended to the Timetable), this section provides further specific details about the course.

COURSE NUMBER : **ECON F213**
COURSE TITLE : **MATHEMATICAL AND STATISTICAL METHODS**
INSTRUCTOR-IN-CHARGE : **PROF. RAHUL ARORA**

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Chamber Number: 6165 D

1. COURSE DESCRIPTION

The course is designed to provide a basic understanding of various statistical and mathematical methods useful for decision-making and for understanding applications in other economics and finance courses. The course is divided into two parts: Statistical Methods and Mathematical Methods. **Statistical Methods** include methods of collecting and describing statistical data – types of data and data collection methodologies, sampling methods & design, data describing methods (graphical and numerical); Correlation and Regression analyses; Index numbers: Concepts, uses, and methods of construction; Analysis of Time-series: Components, data generating process, and stationarity; Statistical Inference: Introduction to Probability and its distributions, sampling distributions, theory of estimation, Tests of Hypotheses testing (Parametric & Non-Parametric); and **Mathematical Methods** include Concepts from Linear Algebra & their Applications (Matrices, Input-Output analysis, and Game Theory); Concepts from Calculus and their applications; Introduction to Optimization in Economics and its Applications.

2. SCOPE AND OBJECTIVE

The main objective is to learn how to use mathematical and statistical methods in economics. It involves a review of the fundamentals of mathematics and statistics, which can be used in applied economics. Knowledge of these fundamentals and their application in applied economics would help students better understand the entire economic system and the policymaking process. At the end of this course, students will be able to understand and solve graduate-level problems in economics and finance using the knowledge they have gained. Specifically, this course uses fundamentals from calculus, algebra, optimization, and basic statistics. This course would lay the foundation for students to understand related courses in economics and finance, such as econometrics and growth theory.

3. COURSE LEARNING OUTCOMES

- CLO1.** Apply appropriate methods for data collection, sampling, organization, and descriptive analysis, using graphical and numerical techniques to summarize and interpret statistical data.
- CLO2.** Analyze relationships among economic and business variables using correlation and regression techniques, and construct and interpret index numbers for relevant applications.
- CLO3.** Analyze time-series data by identifying its components, understanding the underlying data-generating process, and assessing stationarity.
- CLO4.** Apply probability concepts and statistical inference techniques, including probability and sampling distributions, estimation, and parametric and non-parametric hypothesis testing, to draw valid conclusions from data.
- CLO5.** Apply mathematical methods to economic problems, using concepts from linear algebra, input-output analysis, game theory, calculus, and optimization to formulate and solve economic decision problems.

4. TEXTBOOKS AND REFERENCE BOOKS

4.1 Textbook

TBIS - Statistics – Lind, Douglas A., Marchal, William G., and Wathen, Samuel A., *Statistical Techniques in Business and Economics*, McGraw-Hill, 14th International Edition (2010).

TBIM - Mathematics – Alpha. C. Chiang, *Fundamental Methods of Mathematical Economics*,



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McGraw-Hill, 3rd Edition.

4.2 Reference Books

- **RB1**.– David R. Anderson, Dennis J. Sweeney, and Thomas A. Williams, *Statistics for business and economics*, 9th Edition (2005).
- **RB2**. – Ken. Black, *Business Statistics for Contemporary Decision Making*, Wiley India, 4th Edition (2004).
- **RB3**. – S.P. Gupta, *Statistical Methods*, Sultan Chand and Sons, 43rd Edition (2014).
- **RB4**. – Knut Sydsaeter, and Peter T. Hammond, *Mathematics for Economic Analysis*, Pearson Education, 1st Edition (reprint 2002).
- **RB5**: Webster, Thomas J. (2009): Introduction to Game Theory in Business and Economics, Segment Books.

5. Course Format & Weekly Workflow

The course will be conducted using a blended, student-centered, and outcome-based approach, incorporating regular lectures (three per week to study core concepts through usage of blackboard & multimedia tools such as PowerPoint slides, video clippings), tutorial classes (one per week for problem solving and explaining numerical exercises through blackboard/slides/statistical software), and interactive discussions during lectures (to encourage discussion on real-life cases and to promote critical thinking). The students' progress throughout the course will be evaluated through continuous assessment in accordance with the institute's academic regulations. The grading will be relative and in accordance with BITS academic regulations. The course will use NALANDA, a learning management system (LMS), to display course information, including lecture slides/notes, tutorial sheets, solution material, and other related materials. All notices will also be published on NALANDA.

Week No.	Module No. [Description]	Lecture/Tutorial (L/T)	Reference Material	Learning outcome
1	1 [Course Introduction & Methods of collecting and describing data, including Sampling Methods]	L1 : Overview of Statistical and Mathematical concepts to be covered	Course Handout	Understanding of course contents and their significance
		L2–3 / T1–2 : Types of statistics, variables, and levels of measurements; Frequency Table and distribution, their graphic presentation; Numerical Measures to describe data – locational measures and measures of dispersion & Moments; Additional measures of dispersion to show data shape; Sampling methods & Design; Outlier Detection Methods	TB1S : Chapters 1 to 4 & 8; RB3 : Chapters 1 to 9; & Class notes	Knowledge of types of data, their collection, and presentation. Understanding of essential statistical tools and their applications, with the interpretation of the results
2-3	2 [Correlation and Linear Regression]	L4–8 / T3 : Simple correlation, Lag & Lead in correlation; Correlation in time-series including Cross correlation and auto correlation functions; Partial & Multiple correlation analysis; Liner Regression, Multiple Regression analysis	TB1S : Chapters 13 & 14; RB3 : Chapters 10, 11 & 9 (Vol II); & Class notes	Understanding of the basics of correlation and regression, and their applications
4	3 [Analysis of Time-Series]	L9-10 / T4 : Components of a Time-Series: Measurement of Secular Trend, Seasonal Variations, Cyclical Variations, & Irregular	RB3 : Chapter 14; & Class notes	Understanding of basic time-series data handling



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		Variations; Double Structure of a Random Variable, Data Generating Process (DGP), and Stationarity		
5	4 [Index numbers]	L11–14 / T5: Concepts and uses of Index numbers; Construction and Interpretation of weighted and unweighted index numbers	TB1S: Chapter 17; RB3: Chapter 13; & Class notes	Knowledge of constructing Index numbers
Tutorial 6 for Quiz 1 – 8th Sept 2026 (Tuesday)				
6-8	5 [Statistical Inference: Parametric & Non-Parametric]	L15–22 / T7–8: Introduction to various approaches of assigning Probability, Bayes' Theorem, Concepts under Statistical Inference: Steps in hypothesis testing, p-value, level of significance, types of errors; Hypothesis Testing: Parametric: Large sample & small sample tests (z & t) with their applications; One-way & two-way ANOVA and Variance ratio test (F-Test) with their applications; Non-Parametric: Chi-square test & its applications	TB1S: Chapters 5, 8 to 12; 15 & 16 RB3 – Vol II: Chapters 1, 3 to 5, & 11; & Class notes	Understanding of basic concepts of probability, statistical inference, & their applications
9	Assignment Presentations	L23-24/T9: Assignment presentation of 50 % groups based on the topics assigned from the statistical methods	Open Source	Application of concepts in research
Mid-Sem Examination – 08 October 2026 (AN1) Mid-Sem Answer Sheet Distribution in Tutorial 10				
10	6 [Matrix Algebra and its applications, including Input-Output Analysis]	L25–27/T11: Basics of matrices and their usage in economics. Input-Output analysis, including calculation of linkages and multipliers	TB1M: Chapter 4 & 5; & Class notes	Understanding of applications of linear algebra
11-12	7 [Static Game Theory and Its Applications]	L28–33 / T12: Basics of strategic thinking and role of game theory: Types of games & Interaction; Solution Concepts with Static one-time Games played simultaneously: IESDS, Nash Equilibrium, PSNE, MSNE, Odd Rule, Saddle point; Conversion of GTP into LPP; Continuous strategies and their applications to Markets – Cournot, Bertrand, and Stackelberg Models	RB5: Chapters 1-3 & 5; & Class notes	Knowledge of Static Game theory concepts & their applications
Lecture 34 for Quiz 2 – 5th Nov 2026 (Thursday)				
13	8 [Concepts from]	L35–37 / T13: Applications of Simple and partial derivatives to Economics; Applications of	TB1M: Chapters 6 to 9; & Class notes	Understanding of applications of calculus



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	Calculus and their applications]	Maxima and Minima to Economics; Applications of Integration to Economics		
14-15	9 [Optimization techniques and their applications to economics]	L38–40 / T14: Formulation of constrained optimization problem; Constrained optimization – The Lagrange multiplier method; Kuhn-Tucker conditions; Formulation of constrained optimization problem with inequality constraints; Optimization with inequality constraints – Methods in Linear programming; Duality theory of linear programming.	TBIM: Chapters 11 to 13, 18, 19; & Class notes	Knowledge of basics of constrained optimization and its applications
Lecture 42 for Quiz 3 – 26th Nov 2026 (Thursday)				
16	Assignment Presentations	L41 & 43: Assignment presentation of remaining 50 % groups based on the topics assigned from the mathematical methods	Open Source	Application of concepts in research

6. EVALUATION SCHEME

Component	Max Duration	Weightage	Max. Marks	Tentative Dates*	Nature of Component
Assignment & Class Participation@	15 minutes	5%	10	See Weekly Workflow	Open Book
Quiz – 1[#]	45 minutes	15%	30	8 th Sept 2026	Closed Book
Mid-Semester Exam	90 minutes	25%	50	8 Oct 2026 (AN1)	Open Book
Quiz – 2[#]	45 minutes	15%	30	5 th Nov 2026	Closed Book
Quiz – 3[#]	45 minutes	15%	30	26 th Nov 2025	Closed Book
Comprehensive Exam	180 minutes	40 %	80	12 Dec 2026 (FN)	Closed Book

@: The group assignment will be given in groups (a group of six to seven students). Each group will be asked to prepare and give a 15-minute presentation in class.

Best two quizzes out of three would be considered for the final evaluation. All quizzes are announced.

* Quiz Dates are tentative and subject to change as per the circumstances. In case of any change, notice will be published in advance.

Criterion for NC / Minimum passing marks criterion / Minimum performance requirement for a valid grade: A student should obtain 30 % of the average of the top few performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.

7. ATTENDANCE POLICY

Each student is expected to attend all classes and participate in all evaluation components. Discussions during class hours will be highly beneficial for better learning and higher scores. A few more questions (in addition to the numerical in the tutorial sheets) with a greater level of difficulty will be discussed/solved only in the regular and tutorial classes. Please do attend and take advantage of the class discussion. **The AUGSD's attendance policy will be strictly followed.**

8. CHAMBER CONSULTATION HOUR



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Students can approach the Instructor In-Charge (IC) at any time via email. Their queries will be answered via email or in offline meetings, depending on the level of the question. The IC will also be available to meet each Tuesday from 4 PM to 5 PM in his chamber.

9. MAKEUP POLICY

Makeup for the mid-semester exam and comprehensive examination shall be granted only in genuine cases, as per the academic regulations, after a thorough examination of the makeup application. Only those requests would be entertained for a thorough examination who applied **before or within two days of the exam**. Please do not make requests after the application period is over. Those requests would not be entertained. Also, **no makeup request for the quizzes containing one buffer shall be considered.**

10. NOTICES

For course-related announcements, please check your email and LMS (NALANDA) accounts regularly. All the information will also be communicated during the class.

11. NOTE

Quizzes and exams will be conducted during scheduled hours as specified. Changes, if any, will be communicated well in advance. Please check/write your ID number and name before submitting any evaluative component/request for evaluation. Failure to do so may result in unevaluated answer sheets.

INSTRUCTOR-IN-CHARGE
ECON F213